REFLECTION AND REFRACTION OF LIGHT

An Experimental Study of the Laws, Phenomena and Applications of Light

INTRODUCTION

Light is a form of electromagnetic energy that enables us to see objects and observe the world around us. When light encounters the boundary between two media, it may undergo different changes in its path. Two of the most important phenomena associated with the propagation of light are reflection and refraction.

Reflection is the phenomenon in which light returns into the same medium after striking a surface. Refraction, on the other hand, is the change in the direction of light when it passes obliquely from one transparent medium to another.

These phenomena are fundamental concepts in geometrical optics and have numerous applications in everyday life and modern technology. Mirrors, lenses, cameras, telescopes, spectacles, optical fibres and medical instruments all work using the principles of reflection and refraction.

This project investigates these two phenomena through their laws, experimental activities, observations and practical applications.

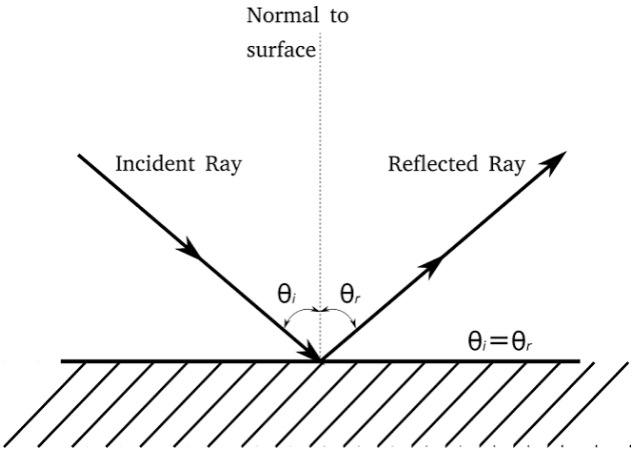
Objectives

- To understand the principles of reflection and refraction.
- To verify the laws of reflection experimentally.
- To study refraction through a glass slab.
- To understand Snell's law and refractive index.
- To study total internal reflection and its applications.
- To relate optical principles to real-life technologies.

REFLECTION OF LIGHT

Reflection is the phenomenon in which a ray of light returns into the original medium after striking a reflecting surface.

When a ray falls on a reflecting surface, it is called the **incident ray**. The ray that returns from the surface is called the **reflected ray**. The perpendicular drawn to the reflecting surface at the point of incidence is called the **normal**.



Laws of Reflection

First Law:

The incident ray, reflected ray and the normal at the point of incidence all lie in the same plane.

Second Law:

The angle of incidence is equal to the angle of reflection.

$$i=r$$

where:

- i= angle of incidence
- r = angle of reflection

Types of Reflection

1. Regular Reflection

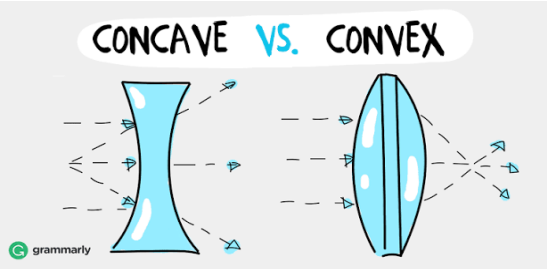
When parallel rays of light fall on a smooth and polished surface, they are reflected in a well-defined direction. This is called regular reflection.

2. Diffuse Reflection

When light falls on an irregular or rough surface, the reflected rays travel in different directions. This is called diffuse reflection.

Although diffuse reflection appears irregular, every individual ray still obeys the laws of reflection.

Spherical Mirrors



Reflection is also used in spherical mirrors.

A **concave mirror** has a reflecting surface curved inward. It can form real or virtual images depending on the position of the object.

A **convex mirror** has a reflecting surface curved outward. It generally produces a virtual, erect and diminished image and provides a wider field of view.

For spherical mirrors:

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

ACTIVITY – VERIFICATION OF THE LAW OF REFLECTION

Aim

To experimentally verify that the angle of incidence is equal to the angle of reflection.

Apparatus Required

- Plane mirror
- White sheet of paper
- Drawing board
- Optical pins
- Ruler
- Protractor

Procedure

1. Fix a white sheet of paper on a drawing board.
2. Place a plane mirror along a straight reference line.
3. Draw a normal to the mirror at the point of incidence.
4. Draw an incident ray making a known angle with the normal.
5. Place two pins along the incident ray.
6. Looking into the mirror, place two additional pins such that they appear to lie on the same straight line as the images of the first two pins.
7. Remove the mirror and join the points to obtain the reflected ray.
8. Measure the angle of incidence and angle of reflection using a protractor.
9. Repeat the experiment for different angles.

Observation

Trial	Angle of Incidence	Angle of Reflection	Difference
1	30°	30°	0°
2	40°	39°	1°
3	50°	49°	1°
4	60°	60°	0°

Observation and Inference

The measured values of the angle of incidence and angle of reflection are nearly equal in every trial. The small differences may occur due to experimental limitations such as parallax, thickness of pins and errors in measuring angles.

Result

The experiment verifies the second law of reflection : $i=r$

APPLICATIONS OF REFLECTION

Reflection is widely used in science and technology.

Rear-view mirrors

Convex mirrors are used as rear-view mirrors in vehicles because they provide a wider field of view.

Reflecting telescopes

Reflecting telescopes use concave mirrors to collect and focus light from distant astronomical objects.

Periscopes

Periscopes use plane mirrors or prisms to change the direction of light and allow objects to be viewed from a different position.

Solar concentrators

Concave reflecting surfaces can concentrate sunlight at a particular point, producing high temperatures useful for solar-energy applications.

REFRACTION OF LIGHT

Refraction is the phenomenon in which the direction of propagation of light changes when it passes obliquely from one transparent medium to another.

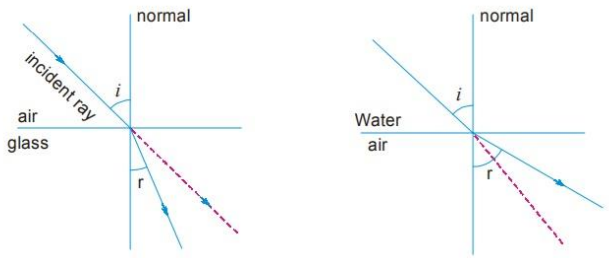
The change occurs because the speed of light is different in different media.

For example, light travels at a different speed in air, water and glass.

The frequency of light remains unchanged when it passes from one medium to another, while its wavelength changes.

The relationship is:

$$v=f\lambda$$



where:

- v = speed of light
- f = frequency
- λ = wavelength

Snell's Law

According to Snell's law:

$$\frac{sini}{sinr} = constant$$

For two given media, this constant is related to the refractive index.

The relative refractive index of medium 2 with respect to medium 1 is:

$$n21 = \frac{sini}{sinr}$$

It can also be expressed as:

$$n21 = \frac{v1}{v2}$$

REFRACTION THROUGH A GLASS SLAB

When a ray of light passes through a rectangular glass slab, it bends towards the normal while entering the glass because glass is optically denser than air.

When it emerges from the glass into air, it bends away from the normal.

Since the opposite surfaces of a rectangular glass slab are parallel, the emergent ray is **parallel to the incident ray**, although it is shifted sideways.

This sideways shift is called **lateral displacement**.

The magnitude of lateral displacement depends on:

- Thickness of the glass slab
- Angle of incidence
- Refractive index of the material

ACTIVITY – DETERMINATION OF REFRACTIVE INDEX OF GLASS

Aim

To determine the approximate refractive index of glass using Snell's law.

Apparatus

- Rectangular glass slab
- White paper
- Drawing board
- Optical pins
- Protractor
- Ruler
- Pencil

Procedure

1. Place the glass slab on a sheet of paper and trace its outline.
2. Draw an incident ray making a known angle with the normal.
3. Place two pins along the incident ray.
4. Observe the images of the pins through the glass slab.
5. Place two more pins so that they appear aligned with the images.
6. Remove the slab and join the points to trace the refracted ray.
7. Measure the angle of incidence and angle of refraction .
8. Calculate:

$$\boldsymbol{n = \frac{sini}{sinr}}$$

9. Repeat for different angles.

Observation Table

Angle of Incidence i	Angle of Refraction	sin i	sin r	$\frac{\sin i}{\sin r}$
30°	19°	0.500°	0.326°	1.53
40°	25°	0.643°	0.423°	1.52
50°	31°	0.766°	0.515°	1.42
60°	39°	0.866°	0.574°	1.51

Observation

The value of $\sin i/sinr$ remains approximately constant for different angles of incidence.

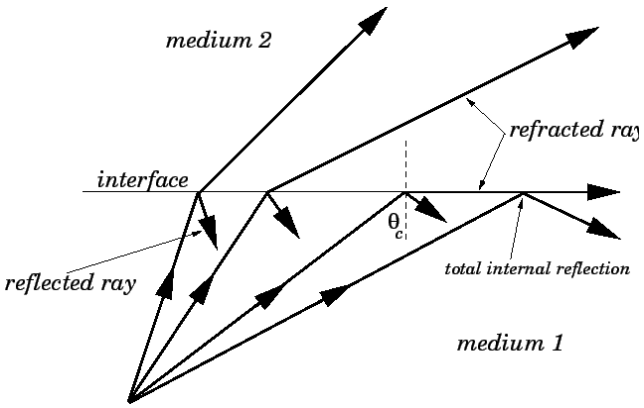
The average value is approximately: **n~1/51**

Result

The experimental value is close to the refractive index of ordinary glass, which is approximately 1.5.

Thus, the experiment supports Snell's law**.**

TOTAL INTERNAL REFLECTION



An important phenomenon associated with refraction is **Total Internal Reflection (TIR)**.

When light travels from an optically denser medium to an optically rarer medium, it bends away from the normal.

As the angle of incidence increases, the angle of refraction also increases.

At a particular angle of incidence, the refracted ray travels along the boundary. This angle is called the **critical angle**.

If the angle of incidence becomes greater than the critical angle, the ray is completely reflected back into the denser medium. This phenomenon is called total internal reflection.

For a medium of refractive index surrounded by air:

$$\sin c = \frac{1}{n}$$

where C is the critical angle.

Conditions for Total Internal Reflection

1. Light must travel from an optically denser medium to an optically rarer medium.
2. The angle of incidence must be greater than the critical angle.

APPLICATIONS OF REFRACTION AND TOTAL INTERNAL REFLECTION

Optical Fibres

Optical fibres transmit light signals through thin glass or plastic fibres using repeated total internal reflection.

They are extensively used in:

- Internet communication
- Telephone communication
- Cable television
- Medical imaging

Endoscopes

Medical endoscopes use optical fibres to transmit images from inside the human body to an external viewing system.

Cameras

Camera lenses use refraction to focus light and form images on an image sensor.

Spectacles

Eyeglasses use lenses to correct vision by controlling the path of light entering the eye.

Microscopes and Telescopes

Lenses in optical instruments use refraction to produce magnified or clearly focused images.

CONCLUSION

The project demonstrates that reflection and refraction are systematic physical phenomena governed by well-defined laws.

The reflection experiment showed that the angle of incidence is approximately equal to the angle of reflection, thereby verifying the law of reflection.

The refraction experiment demonstrated that the ratio remains approximately constant for a given pair of media. This supports Snell's law and gives an approximate refractive index of glass close to **1.5**.

The study of total internal reflection further explains how light can be confined within a medium and transmitted over long distances. This principle forms the basis of optical-fibre communication and several medical technologies.

Thus, reflection and refraction are not only fundamental concepts in geometrical optics but also form the scientific basis of many technologies used in modern life.